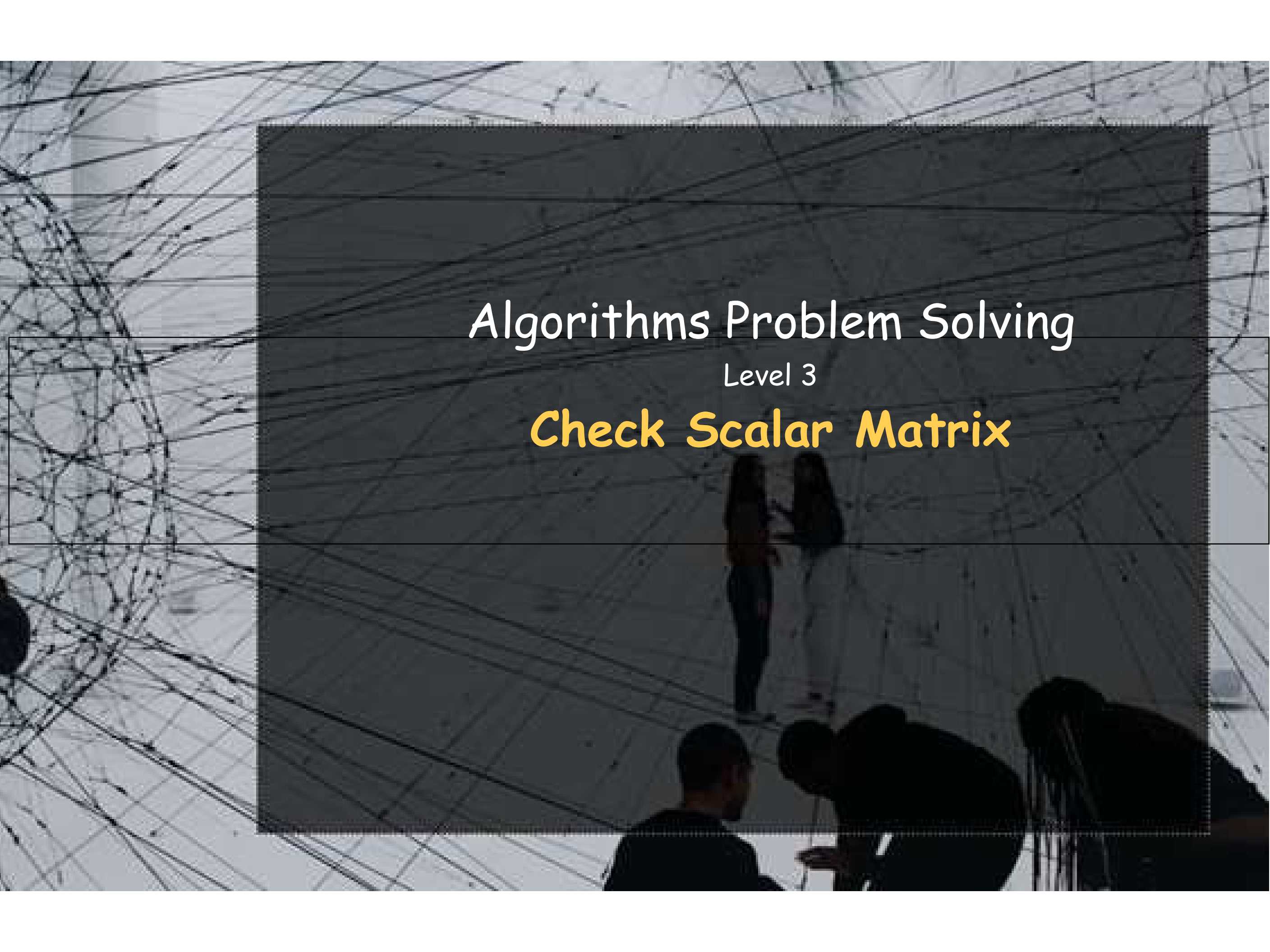


Algorithms Problem Solving

Level 3

Check Scalar Matrix

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the lower part of the image, the silhouettes of several people are visible, some standing and some sitting, suggesting a collaborative workspace or a public area. The overall color palette is muted, with greys, blues, and the warm tones of the light coming through the glass.

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Definition

Scalar Matrix Verification

 Program checks whether a square matrix is a Scalar Matrix.

 A scalar matrix is a diagonal matrix in which all diagonal elements have the same value.

Explanation

Reference Diagonal Value

-  The first diagonal element `matrix[0][0]` is used as the reference scalar value.

-  Every other diagonal element must match this reference.

Diagonal Uniformity

-  The program checks every position where `i == j`.

-  Each diagonal element must equal `matrix[0][0]`.

Off-Diagonal Validation

-  Every position where `i != j` must contain `0`.

-  This guarantees that the matrix is diagonal as well as scalar.

Relationship to Identity Matrix

-  An identity matrix is a scalar matrix whose common diagonal value is `1`.

Procedures

1. Store `matrix[0][0]` in `FirstDiagElement`.
2. Traverse all matrix elements.
3. If `i == j`, compare the value with `FirstDiagElement`.
4. If `i != j`, verify that the value is `0`.
5. Return YES or NO.

💡 Example

matrix1 = { {9,0,0}, {0,9,0}, {0,0,9} };

➡ Result

YES: Matrix is scalar.

Note

- ◆ Every scalar matrix is a diagonal matrix.
- ◆ Not every diagonal matrix is a scalar matrix because diagonal values may differ.
- ◆ The scalar value can be any valid matrix element, including `0`.

⊘ Warning

- ◆ The first diagonal element must be stored correctly before comparing other diagonal elements.
- ◆ All off-diagonal elements must be exactly `0`.
- ◆ Scalar matrices are square matrices, so row and column dimensions must match.



Comparison

`IsScalarMatrix()`

- ✓ Requires all diagonal elements to have the same value.
- ✓ Requires all off-diagonal elements to be `0`.
- ✓ Supports any common diagonal scalar value.

`IsDiagonalMatrix()`

- ✓ Requires only off-diagonal elements to be `0`.
- ✓ Allows different values on the main diagonal.
- ✓ Includes scalar matrices as a special subset.

✦ Summary

- ✓ Uses the first diagonal element as the reference scalar.
- ✓ Verifies equality among all diagonal elements.
- ✓ Confirms that all off-diagonal elements are zero.
- ✓ Demonstrates the relationship between scalar, diagonal, and identity matrices.

Thank you 😊